Sunday, 5/19

6:00 PM - 8:00 PM — Kickoff Reception - Third Floor, League

Monday, 5/20

8:00 AM - 9:00 AM — Breakfast - Concourse/Hussey Room, League

9:00AM - 10:00 AM — Plenary: KT Ramesh - Lydia Mendelssohn Theatre (Located in the League) Chair: Ellen Arruda

10:00 AM - 10:30 AM — Break - Hussey Room

10:30 AM - 12:10 PM — Session 1

	Fluids (CFD) - Lydia Mendelssohn Theatre (Located in the League) Chair: Mauro Rodriguez (confirmed)	Polymer Mechanics - Koessler Room, League Chair: Alan Wineman (confirmed)	Additive Architectured and 2D Materials - Michigan Room, League Chair: Serife Tol (confirmed)
Talk 1	Ying Li, Computational modeling of magnetic particle margination within blood flow throu	Anastasia Muliana, A constitutive model of viscoelastic polymers undergoing microstructural chan	Pu Zhang, A design methodology for 3D auxetic lattice materials with enhanced stiffness and
Talk 2	Ricardo Panesi, Modeling and sensitivity analysis of PEM fuel cells by using CFD technic	Wenlong Li, Thermodynamics response of thermoplastics in the glassy state: Evaluating internal h	Fereshteh A Sabet, 3D printing and finite element simulation of model composites with stiff a
Talk 3	Minki Kim, Inertial collapse of cavitation bubble near an interface	Aayush Prasad, Modelling the mechanics of crystallizable shape memory polymers	Orion L Kafka, A multiscale modeling framework to estimate fatigue life distribution
Talk 4	Harry Lee, Improved upper bounds on the energy dissipation rate for shear flow with inje	Oscar Lopez-Pamies, A general result for the magnetoelastic response of isotropic suspensions o	
Talk 5			

12:10 PM - 2:00 PM — Lunch on your own in Ann Arbor

2:00 PM - 3:40 PM — Session 2

	Fluids (FSI) - Henderson Room, League Chair: Ying Li (confirmed)	Polymer Mechanics - Koessler Room, League Chair: Jon Estrada (confirmed)	Additive Architectured and 2D Materials - Michigan Room, League Chair: Ignacio Arretche (confirmed)
Talk 1	KEYNOTE: Mauro Rodriguez, Rayleigh collapse of a bubble near a soft object	Alexander K Landauer, Porous material deformation characterized by 3D light field imaging	Xiang Zhang, Numerical modeling and design for a 3D printing process based on frontal poly
Talk 2	[Keynote continued]	Stylianos Koumlis, High-strain rate compressive behavior of open-cell polyurethane foams	Anthony N. Palazotto, Evaluation of a celestial structure constructed using additive manufact
Talk 3	Christiana Mavroyiakoumou, Simulating large-amplitude motions of membranes in invisc	Amir Ostadi Moghaddam, An artificial neural network approach to improve the material properties	Joshua Pribe, Fatigue crack growth at weak interfaces: mechanisms relevant for an additive
Talk 4	Sameh Tawfick, Dynamic self-assembly of wet hair	Sagar Vyas, Manufacturing of unidirectional glass-fiber reinforced composites via frontal polymeri	Grace Gu, Structures engineered to mitigate thermally induced stresses in metallic materials
Talk 5		Suzanne Peterson, Modeling of thermal instabilities driven by frontal polymerization of composites	

3:40 PM - 4:10 PM — Break - Hussey Room

4:10 PM - 5:50 PM — Session 3

	Fluids (FSI) - Henderson Room, League Chair: Sameh Tawfick (confirmed)	Damage and Failure of Quasi-Brittle Composites - Koessler Room, League Chair: Marco Salviato, Co-Chair: Anthony Waas (confirmed)	Additive Architectured and 2D Materials - Michigan Room, League Chair: Harley Johnson (confirmed)
Talk 1	KEYNOTE: Rajeev Jaiman, A novel deep learning model for fluid-structure interaction	KEYNOTE: Zdeněk P. Bažant, Design of new materials and structures to maximize strength at pr	Serife Tol, Metamaterial concepts for low-frequency elastic wave focusing
Talk 2	[Keynote continued]	[Keynote continued]	Ying Li, Design of mechanical metamaterials for simultaneous vibration isolation and energy
Talk 3	Vivek Ojha, High-order fluid-structure interaction simulations with adaptive meshing	Avinkrishnan A Vijayachandran, Surrogate-based optimal steered fiber- paths for maximized buck	Ignacio Arretche, Manipulating vibration response of 3D printed metastructures through lattic
Talk 4		Marco Salviato, Fracturing behavior and size effect of discontinuous fiber composite structures	Anastasia Muliana, Turning planar rigid panels into complex freeform shapes
Talk 5			

Tuesday, 5/21

8:00 AM - 9:00 AM — Breakfast - Concourse/Hussey Room, League

9:00AM - 10:00 AM — Plenary: Nadine Aubry - Ballroom, League - Chair: Eric Johnsen

10:00 AM - 10:30 AM — Break - Hussey Room

10:30 AM - 12:10 PM — Session 1

	Fluids (Cavitation and High Rate) - Henderson Room, League Chair: Jon Estrada (confirmed)	Damage and Failure of Quasi-Brittle Composites - Koessler Room, League Chair: Paul Davidson, Co-Chair: Dianyun Zhang (confiirmed)	Additive Architectured and 2D Materials - Michigan Room, League Chair: Grace Gu (confirmed)
Talk 1	KEYNOTE: Fernando Pereira Duda, The interaction between mechanics and fluid perme	KEYNOTE: Samit Roy, Investigating anomalous fracture mechanisms at the nanoscale in quasi-b	Kisik Hong, Cracking and spalling of the oxide layer developed in high-burnup zircaloy-4 clad
Talk 2	[Keynote continued]	[Keynote continued]	Wei Lu, Characterizing the interaction in atomically thin layered materials
Talk 3	Kazuya Murakami, Non-spherical analysis of laser-induced bubble collapse in soft matte	Bin Li, A gradient damage model for polydisperse polymer networks	Sameh Tawfick, The route towards graphene composites
Talk 4	Aditya Kumar, The poker-chip experiment explained	Marco Salviato, Strength and cohesive behavior of thermoset polymers at the microscale: a size-e	Harley T. Johnson, Moire mechanics in 2D materials
Talk 5		Royan John D'Mello, Effect of size and randomness in matrix properties on mechanical response	Shelby Hutchens, Toward an intrinsic cutting energy

12:10 PM - 2:00 PM — Lunch - Ballroom, League

2:00 PM - 3:40 PM — Session 2

	Fluids (Cavitation and High Rate) - Henderson Room, League Chair: Shelby Hutchens (confirmed)	Damage and Failure of Quasi-Brittle Composites - Koessler Room, League Chair: Royan John D'Mello, Co-Chair: Marco Salviato (confirmed)	Data-Driven Modeling - Michigan Room, League Chair: Phani Motamarri (confirmed)
Talk 1	KEYNOTE: Jin Yang, Strain hardening effects of soft viscoelastic materials at extreme lo	Ramesh Talreja, Defect severity quantification for failure in unidirectional composites under transv	Cheng Huang, Multi-fidelity Modeling of Combustion Dynamics
Talk 2	[Keynote continued]	Weijia Chen, Progressive failure prediction of a curved composite flange	Sarah Beetham, Modeling turbulent flows using sparse regression with embedded invariance
Talk 3	Sam Pellone, Vorticity dynamics of shock-driven interfaces	Dandan Wang, Damage mechanisms for nonwovens in textiles	Zhenlin Wang, System identification of the partial differential equations governing pattern-for
Talk 4	Enayat Mahajerin, Prediction of shock levels to packaged products cushioned with foam	Paul Davidson, Multiscale fatigue modeling of composites	Bikash Kanungo, Exact exchange-correlation potentials from electron densities
Talk 5	Michael Wadas, Evolution of perturbed interfaces subjected to large rarefactions	Giorgia Scetta, Crack propagation behaviour of polyurethane thermoplastic elastomers in cyclic fa	Xiaoxuan Zhang, Prediction of effective material properties in multi-component crystalline sol

3:40 PM - 4:10 PM — Break - Hussey Room

4:10 PM - 5:50 PM — Session 3

	Fluids (Cavitation and High Rate) - Henderson Room, League Chair: Jin Yang (confirmed)	Mechanics of Fracture and Failure in Solids - Koessler Room, League Chair: Krishnaswa Ravi-Chandar (confirmed)	Rescheduled Talks - Michigan Room, League
Talk 1	KEYNOTE: Shelby Hutchens, High-speed needle-mediated cavitation	Ahmed Elbanna, Quasi-continuum analysis of networked materials	Trisha Sain, A diffusion coupled constitutive model for thermo-oxidation behavior in polymers
Talk 2	[Keynote continued]	Wai Ching, An adaptive ensemble hybridization of phase field models for brittle and ductile failure	*(rescheduled from Monday Ssession 1 Talk 4, in Polymer Mechanics)*
Talk 3	Jonathan R. Sukovich, A comparative study of the dynamics of laser and acoustically ge	Masoud Behzadinasab, Dynamic ductile fracture characterization with peridynamics: A Sandia Fra	Susanta Ghosh, Computational and analytical modeling for the fracture of micro-architecture
Talk 4	Lauren Mancia, Modeling ultrasound-induced single bubble cavitation in water and agaro	Zhaoxu Meng, Effect of film topology on the impact resistance of nanocellulose films with helicoid	*(rescheduled from Monday Session 3 Talk 5, in Damage and Failure of Quasi-Brittle Composites)*
Talk 5	Sarah Bentil, Stress relaxation response of shock wave impacted brain tissue	Yue Fan, Coupling of thermal and mechanical behaviors: an energy landscape perspective	

Wednesday, 5/22

8:00 AM - 9:00 AM — Breakfast - Concourse/Hussey Room, League

9:00AM - 10:00 AM — Plenary: Michael Thouless - Ballroom, League - Chair: Alan Wineman

10:00 AM - 10:30 AM — Break - Hussey Room

10:30 AM - 12:10 PM — Session 1

	Biomechanics - Henderson Room, League Chair: Rhima Coleman (confirmed)	Mechanics of Fracture and Failure in Solids - Koessler Room, League Chair: Ahmed Elbanna (confirmed)	Modeling and Computation - Michigan Room, League Chair: Nhung Nguyen (confirmed)
Talk 1	David S. Li, A New Robust 3D Constitutive Model for the Passive Properties of Left Vent	Rui Huang, Poroelastic effects on fracture of polymer gels	Victor Lefevre, Homogenization of elastic dielectric composites with rapidly oscillating passiv
Talk 2	David Nordsletten, Viscoelastic Model of Human Myocardium	K. Ravi-Chandar, Characterization of the fracture behavior of poroelastic hydrogels	Rama Bhat, Alternate exact analysis for vibration of beams using novel boundary characteris
Talk 3	Will Zhang, Analyzing the role of viscoelasticity in the residual stress in soft tissues: a ca	Qiming Wang, Self-healing mechanics of polymers	Mateus Forcelini, Application of a poromechanical model to the interpretation of field vane te
Talk 4		Philippe Geubelle, Transverse failure of fiber-reinforced composites: sensitivity on microstructural	Augusto Bopsin Borges, Effects of residual stresses on bored cast in situ piles capacity - com
Talk 5		Prajwal Arunachala, Strain induced crystallization and fracture of rubber-like materials	Victor Lefevre, A simple explicit homogenization solution for the macroscopic elastic respons

12:10 PM - 2:00 PM — Lunch on your own in Ann Arbor

2:00 PM - 3:40 PM — Session 2

	Biomechanics - Henderson Room, League Chair: Amy Wagoner Johnson (confirmed)	Fracture Mechanics - Koessler Room, League Chair: Michael Thouless (confirmed)	Modeling and Computation - Michigan Room, League Chair: Tom Pence (confirmed)
Talk 1	Azadeh Sheidaei, Study the effect of bone microstructure on bone remodeling	Kenneth M. Liechti, Rate-dependent traction-separation relations for a silicon/epoxy interface infor	Ruben Adorno, Optimization of a spherical geometry for vacuum lighter than air applications

Talk 2	Chenxi Yan, Effect of fatigue on the impact response of rat ulna	Jia-Liang Le, Rate-dependent scaling in quasibrittle fracture	Saied Taheri, Multi-length scale modeling of dry/wet traction
Talk 3	Sara G. Moshage, Equine proximal phalanx bone properties during growth	James Gorman, Exploration of non-crack singularities on the single lap-shear joint	Reyolando M.L.R.F. Brasil, Theoretical and experimental analysis of nonlinear geometric stiff
Talk 4	Hyunggwi Song, Changes in murine bone morphology, apparent density, and gait during	Oscar Lopez-Pamies, Fracture and healing of elastomers: a phase-transition theory & numerical i	John Shaw, Viscoelastic Eshelby analysis of single cardiac myocyte contraction in the cell-in-
Talk 5	Moataz Abdulhafez, Mechanics of skull fixation in pediatric neurosurgery: modeling age	Kenneth M. Liechti, A dual actuator mixed-mode interaction tester	
	3:40 PM - 4:10 PM	Break - Hussey Room	
	4:10 PM - 5:50 PM	**Session 3**	
	Biomechanics - Henderson Room, League **Chair: Anastasia Muliana (confirmed)**		**Modeling and Computation - Michigan Room, League** **Chair: Alan Wineman (confirmed)**
Talk 1	KEYNOTE: Amir Ostadi Moghaddam, Characterization of the elastic constants of soft an		KEYNOTE: Thomas J Pence, Swelling induced shearing and twisting in fiber reinforced hype
Talk 2	[Keynote continued]		[Keynote continued]
Talk 3	Callan Luetkemeyer, Known microstructural difference in ACL bundles detected with full-		Mohammad A. Kazemi-Lari, Axisymmetric and asymmetric inflation of circular membranes wi
Talk 4	Javad Baqersad, Monitoring of skin using a non-invasive skin diagnosis methodology		Adair Aguiar, Strong ellipticity conditions for orthotropic nonlinear elastic materials in plane st
Talk 5	Justin Scott, Materials properties of the thigh in the seated and prone positions		
	6:00 PM - 7:00 PM	Reception - Hussey, League (Must Purchase a Ticket to Attend)	
	7:00 PM - 9:00 PM	Banquet - Hussey, League (Must Purchase a Ticket to Attend)	
	Thursday, 5/23		
	8:00 AM - 9:00 AM	Breakfast - Concourse/Hussey Room, League	
	9:00 AM - 10:40 AM	**Session 1**	
	Biomechanics - Henderson Room, League **Chair: Callan Luetkemeyer (confirmed)**	**Plasticity - Koessler Room, League** **Chair: Oscar Lopez-Pamies (confirmed)**	**Modeling and Computation - Michigan Room, League** **Chair: Reyolando Brasil (confirmed)**
Talk 1	KEYNOTE: Nikolaos Bouklas, Soft tissue mechanics during self-healing	Charles Lieou, Dynamic recrystallization in adiabatic shear banding: an entropic, effective-temper	KEYNOTE: Veera Sundararaghavan, Solving differential equations on quantum computer
Talk 2	[Keynote continued]	Tao Jin, Incorporation of deformation twinning in single-crystal elasto-viscoplasticity model with en	[Keynote continued]
Talk 3	Ryan Rosario, Shear and transverse characterization of the patellar tendon	Pinar Acar, Multi-fidelity crystal plasticity modeling for titanium-aluminum alloys	Walter J. P. Casas, Optimization of a predefined natural frequency for fluid-structure systems
Talk 4	Mohammad Saber Hashemi, Mechanical homogenization of the liver tissue with a combi	Isha Gupta, Mechanics of oxidation of zirconium alloys in pressurized-water reactors	Amin Ghadami, Model-free forecasting of the flutter speed and post-flutter dynamics
Talk 5	Silas Alben, Dynamics and locomotion of flexible foils in a frictional environment	Nhung Nguyen, Mixed-mode cohesive law with a physical crack propagation consideration	
	10:40 AM - 11:00 AM	Coffee	
	11:00 AM - 12:40 PM	**Session 2**	
	Biomechanics - Henderson Room, League **Chair: Nikolaos Bouklas (confirmed)**	**Plasticity - Koessler Room, League** **Chair: Isha Gupta (confirmed)**	**Battery Materials - Michigan Room, League** **Chair:**
Talk 1	KEYNOTE: Anastasia Muliana, Biomechanical properties of bioenergy sorghum stems a	Iman Javaheri, Component-scale microstructure reconstruction using Markov random fields: valid	KEYNOTE: Wei Lu, Coupling mechanics and electrochemistry for modeling of Li-ion batteries
Talk 2	[Keynote continued]	Aaditya Lakshmanan, Dislocation patterning and pattern transition in fatigue using reaction-advec	[Keynote continued]
Talk 3	Ameneh Maghsoodi, Exposing the injection machinery dynamics of T-even bacteriophag	Srihari Sundar, Classification and data mining approach for deformation process sequence selecti	William LePage, Lithium mechanics: creep as dominant deformation mechanism for battery-r
Talk 4	Kejie Chen, Collective cargo transport on multiple tracks	Lucas Sardinha de Arruda, Theoretical and numerical nonlinear dynamic study of the von Mises s	Gaurav Singh, Delamination study at electrode-binder interface in Lithium Ion Battery
Talk 5	Siamak Attarian, Studies of rupture in abdominal aortic aneurysms	Prithvi Thakur, Magnitude frequency distribution of simulated earthquake sequences in damaged	Vijay Venkatesh, Morphology-dependent mass transport model for mechanoelectrochemistry